AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

Course No: CSE 2103
Course Title: Data Structures Lab

Assignment No: 07

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Question:

Implement BFS and DFS in two separate .cpp files.

Solution Code:

1. BFS:

```cpp
#include<bits/stdc++.h>
using namespace std;

void bfs(vector<vector<int>>& adj, int stNode) {
   vector<bool>visited(adj.size(), false);
   queue<int> q;
   q.push(stNode);

   while(!q.empty()) {
       int front = q.front();
       q.pop();

       if(!visited[front]) {
           cout << front << " ";
           visited[front] = true;

           for(int neighbour: adj[front]) {
               if(!visited[neighbour]) {
                   q.push(neighbour);
               }
           }
       }
   }
}

int main() {
   vector<vector<int>> adj = {{3, 2, 1}, {0, 3}, {0,3}, {0, 1, 2, 4}, {3}};
   bfs(adj, 0);
   cout << "\n";
   return 0;
}
```

2 . DFS:

```
#include<bits/stdc++.h>
using namespace std;

void dfs(vector<vector<int>>& adj, int stNode) {
   vector<bool>visited(adj.size(), false);
   stack<int> s;
   s.push(stNode);

   while(!s.empty()) {
       int top = s.top();
       s.pop();

       if(!visited[top]) {
           cout << top << " ";
           visited[top] = true;

           for(int neighbour: adj[top]) {
               if(!visited[neighbour]) {
                   s.push(neighbour);
               }
           }
       }
   }
}

int main() {
   vector<vector<int>> adj = {{3, 2, 1}, {0, 3}, {0,3}, {0, 1, 2, 4}, {3}};
   dfs(adj, 0);
   cout << "\n";
   return 0;
}
```

Sample Input Output:

1. **BFS Input Output:**

```
vector<vector<int>> adj = {{3, 2, 1}, {0, 3}, {0,3}, {0, 1, 2, 4}, {3}};
```

```
0 3 2 1 4
```

2. **DFS Input Output:**

```
vector<vector<int>> adj = {{3, 2, 1}, {0, 3}, {0,3}, {0, 1, 2, 4}, {3}};
```

```
0 1 3 4 2
```